SKITII PRODUCT DOCUMENTATION

Music Ranking & Category Guide

How Skitii chooses music categories, ranks tracks, learns from each session, and personalizes playback for every patient.

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1. What Skitii Is Trying to Do

Most music apps play a fixed playlist. Skitii is different.

Skitii listens to the patient's body signals during a session — mainly **heart rate (HR)** and **heart rate variability (RMSSD)** — and uses that information to:

1. **Pick the right type of music** (calming, grounding, etc.)
2. **Pick the best track** within that type
3. **Learn over time** which tracks actually help that patient

The goal is simple: **play music that supports the patient's current physiological state**, and **get better at it with every session**.

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2. Key Ideas (Simple Definitions)

Term	Plain meaning
Heart Rate (HR)	How fast the heart is beating (beats per minute). Higher often means more stress or activation.
RMSSD	A measure of heart rate variability (shown as **HRV** in the Admin panel). Higher RMSSD often means the body is more relaxed and recovering.
Phase	One measurement window in a session (e.g. a few minutes of monitoring).
HRM Cycle	One full heart-rate monitoring cycle within a session. Cycle 1 uses HR; later cycles use RMSSD.
Category (Music Tier)	A group of music with a similar emotional/physiological purpose: Grounding, Calming, Restorative, or Uplifting.
Track	One individual music file (one song).
Ranking	A score attached to each track. Higher score = "this track is more likely to be chosen first."
Personalization	When the system has learned enough about a specific patient to use *their* rankings instead of global ones.

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3. The Four Music Categories

Skitii organizes all music into **four tiers**. Think of them as four "moods" or "support levels":

```
Most stressed / lowest recovery          Most relaxed / highest recovery
        ↓                                                ↓
   Grounding  →  Calming  →  Restorative  →  Uplifting
```

What each category means

Category	When it is typically used	Patient state (simplified)
Grounding	High stress, very low recovery	Heart racing, body not settling
Calming	Elevated stress, needs settling	Activated but not extreme
Restorative	Moderate recovery	Body starting to balance
Uplifting	Good recovery, relaxed state	Calm, open, positive energy

Category conditions (configured in Admin)

Each category is defined in **Music Therapy** → **Categories** with three ranges. These are the active conditions in Skitii Admin:

Category	HR (bpm)	HRV / RMSSD (ms)	Music BPM
Grounding	95 – 300	0 – 20	50 – 60
Calming	85 – 94	21 – 35	61 – 70
Restorative	70 – 84	36 – 55	71 – 80
Uplifting	0 – 69	56 – 100	81 – 90

How to read this table:

- **HR (bpm)** — heart rate range that points to this category during the **first session cycle**.
- **HRV / RMSSD (ms)** — heart rate variability range that points to this category from **cycle 2 onward**.
- **Music BPM** — the tempo band for tracks in that category (beats per minute).

> **Note:** In the live session app, HRV is measured as **RMSSD**. The Admin panel label is **HRV**; both refer to the same signal used for category matching.

Each track in the library belongs to one of these categories. Individual tracks can also have **trigger ranges** — specific HR or RMSSD values where that track is especially suitable.

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4. How the Category Is Chosen During a Session

Music selection happens in **two stages** across the session.

Stage A — First cycle: based on Heart Rate (HR)

At the **start of the session** (HRM Cycle 1), Skitii looks at the patient's **average heart rate** and maps it to a category using the **HR** ranges from Admin:

Average HR	Category chosen
95 – 300	Grounding
85 – 94	Calming
70 – 84	Restorative
0 – 69	Uplifting

Example: If baseline HR is 65 → **Uplifting** music is chosen for the opening.

This gives a sensible starting point based on how activated the patient's body is.

Stage B — Later cycles: based on RMSSD (adaptive)

From **Cycle 2 onward**, Skitii switches to **RMSSD (HRV)** because it better reflects recovery and nervous system balance, using the **HRV / RMSSD** ranges from Admin:

RMSSD (HRV) value	Category chosen
0 – 20	Grounding
21 – 35	Calming
36 – 55	Restorative
56 – 100	Uplifting

Example: If RMSSD is 35.94 → **Restorative** music.

The "20% change rule" — when to change music

After the first RMSSD reading, Skitii compares each new RMSSD value to the **previous one** and asks:

> "Did the body change meaningfully since last time?"

RMSSD change	What happens
20% or less	Keep the **same category** and keep the **same track**. The body is relatively stable — no need to disrupt the music.
More than 20%	Re-evaluate. The body shifted enough that music may need to adapt.

Why 20%?
Small natural fluctuations happen all the time. The 20% threshold prevents the music from jumping around on tiny changes. Only meaningful shifts trigger a change.

Example from a real session pattern:

- Phase 2: RMSSD drops 8.88% → **same track** (under 20%)
- Phase 4: RMSSD jumps 47.33% → **significant change** → music may progress to next ranked track
- Phase 6: RMSSD changes only 2.61% → **same track** again

What happens when change is over 20%

Two things can occur:

1. **Category changes** → Skitii moves to a new pool of tracks in the new category (e.g. from Calming to Restorative).
2. **Category stays the same** → Skitii stays in the same category but may **pick a different track** — moving down a ranked list of best tracks for that tier.

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5. How a Specific Track Is Chosen (Ranking Logic)

Once the category (or trigger match) defines a **pool of candidate tracks**, Skitii picks one track using **ranking**.

Step 1 — Build the candidate pool

Skitii looks for tracks in this order:

1. **Trigger match** — tracks whose HR or RMSSD range exactly fits the current reading (most precise match).
2. **Category match** — all tracks in the chosen tier (Grounding, Calming, etc.).
3. **Fallback** — if nothing matches, use Calming or any available track.

When RMSSD change is **small (≤ 20%)**, Skitii **only** looks within the current category — it will not pull a track from another category even if a trigger would match.

Step 2 — Sort tracks by ranking (highest first)

Every track has a **ranking score**. Higher score = played sooner.

Tracks are sorted from **best to worst**. The system then picks based on session progression:

Situation	Which track plays
First cycle (HR phase)	Top-ranked track in the pool (position 1)
RMSSD change ≤ 20%	**Same track as before** — no change
RMSSD change > 20%	Move to the **next position** in the ranked list (position 1, then 2, then 3...)

Example:
If Restorative has 3 tracks ranked A (score 20), B (score 8), C (score 6):

- First significant RMSSD event → Track A
- Next significant event (same category) → Track B
- Next → Track C

This creates a **gentle progression** through proven tracks rather than random switching.

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6. Two Types of Ranking: Global vs Personalized

Skitii maintains **two layers of learning**.

Global ranking (for everyone)

- Every track starts with a base score (default: 0).
- After each session phase (from Phase 2 onward), Skitii records how the patient's RMSSD changed while that track was playing.
- The ranking moves up or down:

RMSSD direction after the phase	Ranking change
Improved (positive change)	**+1**
Worsened (negative change)	**−1**
No meaningful direction	**0**

Important: The system cares about **direction** (better or worse), not the exact percentage. A 5% improvement and a 50% improvement both add +1.

Over time, tracks that consistently help patients rise to the top. Tracks that don't help sink lower.

This is **crowd wisdom** — the library learns from all patients.

Personalized ranking (for one patient)

The same +1 / −1 / 0 logic also runs **per patient, per track**.

But personalization only **takes over** when the patient has enough history:

> **Personalization is enabled when the patient has personalized data for at least 5 different tracks in every active music category.**

Until then, Skitii uses **global rankings**.

Once enabled:

- Only tracks with personalized records are used.
- Rankings reflect **what works for this specific patient**, not the average user.
- The screen shows **"P"** (personalized) instead of **"G"** (global).

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7. What the User Sees on Screen (Indicators)

During live monitoring, small badges help staff understand the music logic:

Badge	Meaning
G	Using global (population-wide) ranking
P	Using personalized ranking for this patient
R1, R2, R3...	Track position in the global ranked list (R1 = best)
PR1, PR2...	Track position in the personalized ranked list
12% (example)	How much RMSSD changed between phases
SAME	Change was ≤ 20% → same track will continue
NEXT	Change was > 20% → system will move to next ranked track

Green/red coloring on the percentage shows whether the change crossed the 20% threshold.

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8. End-to-End Session Flow (Story Example)

Imagine a patient starting a session with HR = 65:

Cycle 1 — HR phase

- Category: **Uplifting** (HR below 70)
- Pool: all Uplifting tracks
- Track: highest-ranked Uplifting track (R1)

Cycle 2 — First RMSSD = 35.94

- Category: **Restorative** (RMSSD in the 36–55 band)
- First RMSSD cycle → pick top Restorative track
- Significant change gate: first cycle always applies ranking

Cycle 3 — RMSSD = 32.75 (drops ~8.9%)

- Change < 20% → **same Restorative track**
- Category unchanged

Cycle 4 — RMSSD = 48.56 (jumps ~47%)

- Change > 20% → **next ranked Restorative track**
- Still Restorative category

After session — learning

- For each phase from Phase 2 onward, the system saves:
  - Which track played
  - Whether RMSSD improved or worsened
- Global ranking updates for that track
- Patient's personal ranking updates too

After many sessions

- Patient has 5+ personalized tracks in each category
- Personalization turns on
- Future sessions use **their** rankings (PR1, PR2...) instead of global ones (R1, R2...)

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9. Why This Approach Makes Skitii Stronger Than Typical Products

Here is how Skitii compares to common alternatives:

Approach	Limitation	How Skitii is different
Fixed playlist apps (Spotify, Calm, etc.)	Same music for everyone, no body feedback	Skitii reacts to live HR and RMSSD
Manual clinician choice	Depends on judgment each time; hard to scale	Skitii uses consistent, repeatable rules
Simple "if HR high → play slow music"	One rule, no memory, no learning	Skitii has tiers, triggers, 20% stability gate, and ranked progression
Biofeedback without learning	Adapts in the moment but forgets next session	Skitii **remembers** which tracks helped (+1/−1 over time)
Personalization without enough data	Can overfit too early	Skitii waits until **5 tracks per category** before switching to personal mode
Frequent track switching	Can feel jarring and break focus	The **20% rule** keeps music stable when the body is stable

Product quality strengths in one sentence each

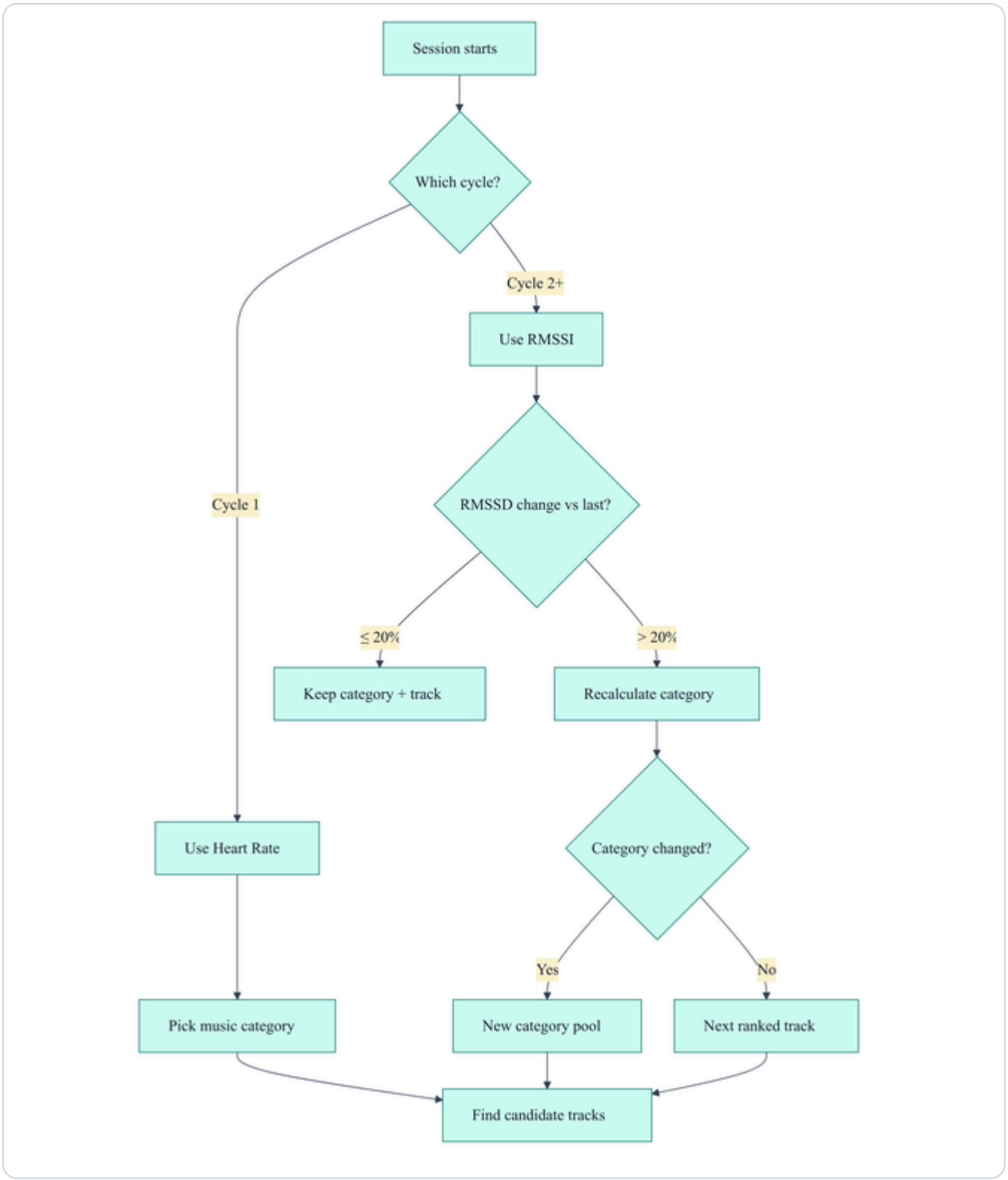
1. **Physiologically grounded** — Music type matches what the body is actually doing, not just a generic “relaxation” label.
2. **Stable when it should be** — Small body fluctuations do not cause unnecessary music changes.
3. **Responsive when it matters** — Large shifts (>20%) trigger meaningful adaptation.
4. **Gets smarter over time** — Every session teaches the system which tracks work.
5. **Personal when ready** — Early sessions use proven global data; later sessions become tailored to the individual.
6. **Transparent for clinicians** — On-screen badges (G/P, R1, NEXT/SAME, %) show *why* a track was chosen, building trust.
7. **Structured music library** — Four clear tiers plus per-track trigger ranges give precision without chaos.

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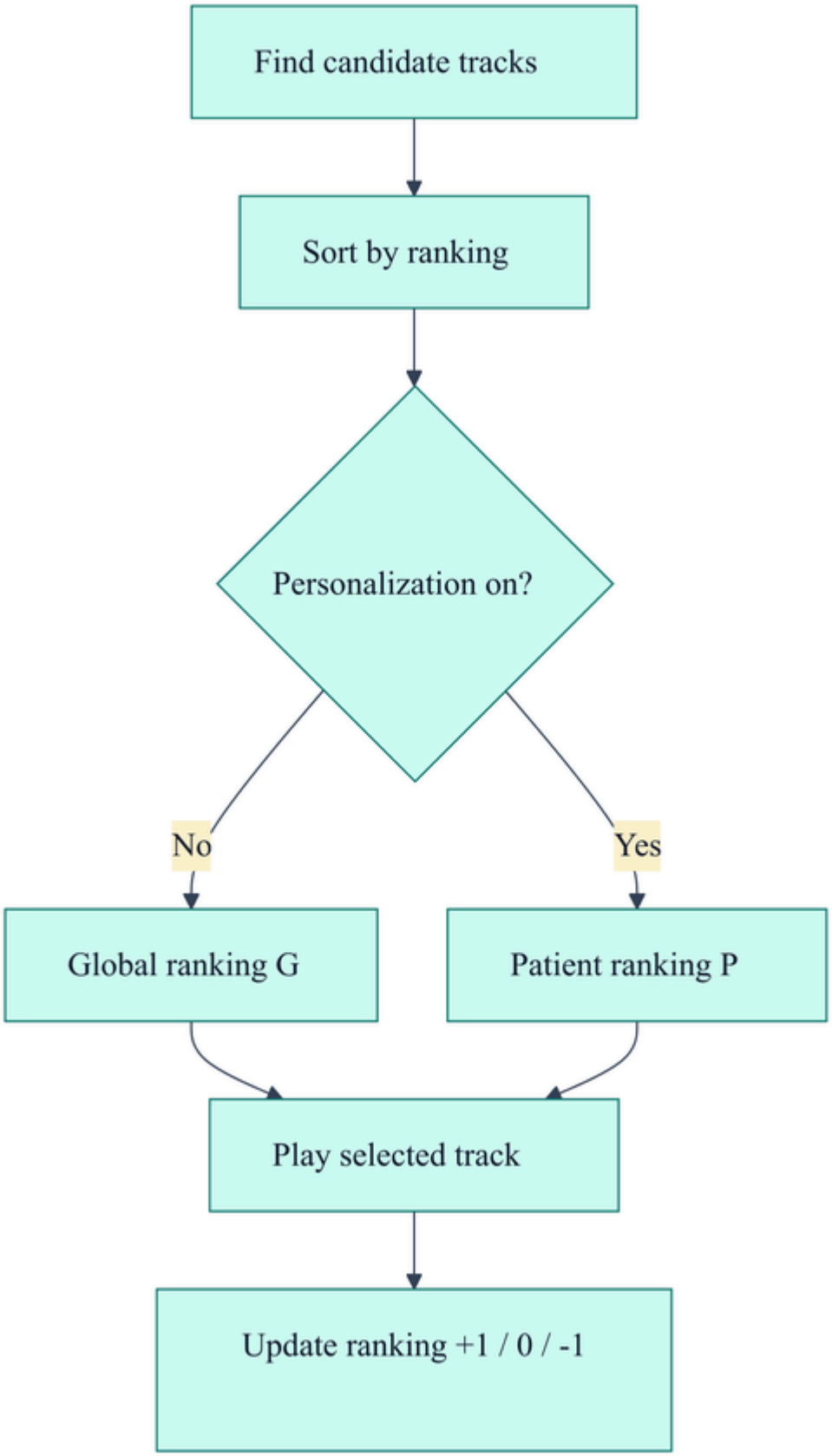
10. Simple Diagram — Decision Flow

The flow is split into two charts for readability: category selection, then track ranking.

10.1 Category selection



10.2 Track ranking and playback


Find candidate tracks
Sort by ranking
Personalization on?
No
Yes
Global ranking G
Patient ranking P
Play selected track
Update ranking +1 / 0 / -1

11. Frequently Asked Questions (Non-Technical)

Q: Why doesn't Skitii change music every time RMSSD moves a little?

A: Because small changes are normal. The 20% rule avoids restless switching and lets the patient settle into the music.

Q: What if a patient is new?

A: Global rankings from all users guide track selection until the patient has enough session history.

Q: Can a Calming track play when Restorative was chosen?

A: Only in specific cases — when RMSSD change is large (>20%) and a track's **trigger range** exactly matches the current reading. During stable periods, Skitii stays within the chosen category.

Q: Does a bigger RMSSD improvement give more ranking points?

A: No. Only the **direction** matters: better = +1, worse = −1. This keeps the system simple and fair across different patients.

Q: What happens if ranking is tied?

A: Tracks are ordered consistently (by a stable internal ID) so the same input always gives the same order.

Q: Is the music choice random?

A: No. It follows clear rules: body signal → category → ranked list position → learning from outcomes.

12. Summary

Layer	What it does
Categories	Match music type to body state (Grounding → Uplifting)
Triggers	Fine-tune track choice to specific HR/RMSSD ranges
20% gate	Prevent unnecessary music changes
Ranking	Order tracks from most to least effective
Global learning	Improve the library for all patients
Personal learning	Tailor rankings to one patient after enough data
Progression	Move through ranked tracks when the body shifts significantly

Skitii is not just “play relaxing music.” It is a **closed-loop system**: the body informs the music, the music is evaluated by how the body responds, and future sessions benefit from that feedback.